

Skills

Languages: Python, R, SQL, TypeScript, JavaScript

ML/Data: PyTorch, OpenCV, Roboflow, SAM2, PaddleOCR, scikit-learn, Pandas, NumPy, SciPy, statistical modeling, Elo systems, Monte Carlo simulation

Apps/Infra: R Shiny, RMarkdown, FastAPI, Next.js, React, PostgreSQL, Supabase, Drizzle ORM, Streamlit, Recharts, Plotly, Grafana, Power BI, Git

AI Engineering: LLM application design, tool/function calling, MCP concepts, RAG patterns, agent workflows, structured outputs, model evaluation

Experience

STARS Advanced Radiology Services

Grand Rapids, Michigan (Remote)

Data Analyst

04/2022 - Present

- Own an analytics portfolio across billing, physician productivity, scheduling, and clinical operations, supporting ~100 administrators and 200+ physicians through 6 dashboards, 20+ reports, and 20+ automated workflows.
- Built and maintain a multi-tab R Shiny billing application end to end, from SQL transformations and KPI design through training and enhancements; used daily by 25-30 employees and executives as Days A/R fell from above 60 to 29-32.
- Replaced a manual query-to-Excel workflow with an interactive Critical Results dashboard that prioritizes urgent physician support requests; compliance improved from ~89% to above 98%.
- Contributed to a Python regression model that evaluates modality, RVU workload, and RVUs per FTE every 2 minutes, enabling surge warnings and data-driven on-call staffing.

Byrne Electrical Specialists Inc.

Rockford, Michigan

Data Science Intern

(Summers 2019 - 2021)

- Built PostgreSQL-backed IoT sensor pipelines, Grafana/Power BI dashboards, and SPC alert logic for robotic assembly OEE, contributing to a 15% improvement in machine OEE.

Selected Projects - live demos at bendotwillcox.github.io/portfolio

ValoMapped - Esports Elo Analytics Platform | Next.js, TypeScript, PostgreSQL, Drizzle ORM, Recharts

- Built and deployed a live Valorant Champions Tour analytics platform with automated ingestion, map-specific Elo ratings, match predictions, player ratings, record books, pick/ban analysis, grid-searched rating calibration, and Monte Carlo tournament simulations.

HoopStats - AI Basketball Video Analytics Pipeline | Python, PyTorch, OpenCV, Roboflow, SAM2

- Implemented video I/O, RF-DETR detection, ByteTrack/SAM2-style tracking, team color classification, court homography, trajectory normalization, and early CSV export scaffolding for a broadcast basketball analytics pipeline.

Voting Paradigm - Voting Systems and Social Choice Demos | Python, FastAPI, Next.js, TypeScript, Supabase

- Built a hosted social-choice demo portfolio with voting-method comparison, apportionment/districting, preference-agent replay, and a tested Python package for plurality, approval, IRV, Borda, Ranked Pairs, score, and quadratic voting.

Education

Michigan State University - B.S. Economics, Minor in Quantitative Data Analytics | Dec 2021